IEEE CASS Internship 2026

Digital Energy Meter

Project Report

Abstract

A digital energy meter was designed using Verilog HDL with an FSM-based architecture. The project was simulated, synthesized, and implemented through the complete ASIC physical design flow.

Introduction

Digital energy meters provide accurate energy measurement using digital hardware. This project demonstrates RTL design, verification, synthesis, and ASIC implementation.

Objectives

• Design using Verilog HDL • Verify using simulation • Complete ASIC flow • Analyze energy accumulation

Design Methodology

The design progressed through RTL coding, simulation, synthesis, floorplanning, power planning, placement, CTS, routing, and signoff.

Results

Functional verification and ASIC implementation stages completed successfully.

Conclusion

The project strengthened practical knowledge of digital design, Verilog, and ASIC implementation.

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